

D-MorphNet

Face Morphing Detection — EfficientNet-B6 + SVM

Results Analysis, Model Problems, and Improvement Plan

89.9%

TEST ACCURACY

0.965

ROC-AUC

0.90

F1-SCORE

54

MISSED MORPHS (FN)

Prepared as a technical review of the reported model performance. This document analyzes results only and proposes changes; it does not modify the training code.

1. Executive Summary

The D-MorphNet system detects face-morphing attacks using a two-stage design: a frozen EfficientNet-B6 backbone extracts a 2304-dimensional deep feature vector from each face, and a Support Vector Machine (SVM) makes the final real-vs-morph decision. On the held-out test set the reported system reaches **89.9% accuracy**, **0.90** precision / recall / F1, and a strong **ROC-AUC of 0.965**.

These are good results for a compact, data-efficient pipeline. However, the confusion matrix and the architecture-comparison table reveal several weaknesses that limit real-world reliability — most importantly a **10.1% overall error rate**, a relatively high **false-positive rate (14.8% of real faces flagged as morphs)**, and **54 morphs that slip through undetected**, which is the most dangerous error type in a biometric setting. This report quantifies each problem and gives a prioritized, concrete plan to improve the model.

Reported results at a glance

Metric	Value	Reading
Accuracy	89.9%	1 in 10 predictions is wrong
Precision (morph)	0.90	10% of morph alarms are false
Recall (morph)	0.90	10% of real morphs are missed
F1-score	0.90	balanced but capped ~0.90
ROC-AUC	0.965	excellent ranking / separability
False Negatives	54 / 1000	undetected morphs (critical)
False Positives	148 / 1000	real faces wrongly flagged

2. Confusion Matrix Analysis

The test set contains 1,000 real and 1,000 morphed images. The reported confusion matrix is:

	Predicted Real	Predicted Morph
Actual Real	852 (TN)	148 (FP)
Actual Morph	54 (FN)	946 (TP)

Green = correct, orange = false positive, red = false negative (the critical error).

2.1 What the errors mean

- **True Negatives (852):** real faces correctly accepted.
- **True Positives (946):** morphs correctly caught — a solid 94.6% detection rate.
- **False Positives (148):** real faces wrongly flagged as morphs. Tolerable in security (a human reviewer can re-check), but 14.8% is high enough to create operational friction at a border gate or enrollment desk.
- **False Negatives (54):** morphs accepted as genuine. **This is the most dangerous outcome** — an undetected morph lets two people share one identity document, which is the exact fraud the system exists to stop.

2.2 The core imbalance

The system is tuned toward catching morphs (high recall) but pays for it with false positives: FP (148) is nearly 3x FN (54). In a real deployment the acceptable trade-off depends on the use case — a passport-issuance pipeline should push FN as close to zero as possible even at the cost of more manual review, whereas a high-throughput gate cannot afford 15% of travelers being stopped. The current single fixed threshold does not let operators express that choice.

3. Architecture Comparison (Table 2)

The reported comparison shows why EfficientNet-B6 was chosen, and also exposes how sensitive the whole system is to the feature extractor:

Model	Classifier	Feat. dim	Acc. %	Prec.	Rec.	F1
EfficientNet-B0	Softmax	1280	62.4	0.63	0.61	0.62
EfficientNet-B5	SVM (Linear)	2048	66.13	0.66	0.66	0.66
EfficientNet-B6 (Proposed)	SVM (RBF)	2304	89.9	0.90	0.90	0.90

The red flag: B0 and B5 score only 62–66%, barely above the 50% random baseline, while B6 jumps to 89.9%. A 23-point gap between B5 and B6 is unusually large and suggests the classifier is **fragile** — the SVM is not learning a robust morph signal, it is riding entirely on whatever extra capacity B6 provides. A truly robust detector would degrade gracefully across backbones, not collapse. This is the strongest evidence that the pipeline is under-fitting the morph problem and over-relying on one frozen feature extractor.

4. Model Problems

The following weaknesses are visible in the reported results and in the design itself.

P1. High false-positive rate (14.8%)

HIGH

Almost 1 in 7 genuine faces is flagged as a morph. At scale this floods human reviewers and erodes trust in the system.

Evidence: FP = 148 / 1000 real images.

P2. Undetected morphs / false negatives

CRITICAL

54 morphs are accepted as real. In morph-attack detection the cost of a false negative is far higher than a false positive, yet the system is not explicitly optimized to minimize it.

Evidence: FN = 54 / 1000 morphs; recall capped at 0.946 on TP count.

P3. Fragile dependence on the backbone

HIGH

Accuracy collapses from 89.9% (B6) to 66% (B5) and 62% (B0). The SVM adds little robustness of its own; the result hinges on one frozen network, so any domain shift that hurts B6 features breaks the system.

Evidence: Table 2: 23-point B5-to-B6 gap.

P4. Frozen features, no morph-specific learning

MEDIUM

The backbone is trained on ImageNet object classes, not on morphing artifacts. Global average pooling discards the fine local cues (blending seams, ghosting near eyes/nostrils/hairline) that best reveal morphs. The model never learns what a morph specifically looks like.

Evidence: Design: include_top=False + GAP, backbone frozen by default.

P5. Single morphing method / weak generalization

HIGH

Training morphs are generated by one landmark-based technique. A detector trained on one morph family typically fails on unseen ones (GAN morphs, other tools, print-scan re-digitization). This is the well-documented cross-morph generalization gap in the literature.

Evidence: Dataset built from a single morph generator.

P6. Single fixed decision threshold

MEDIUM

One global threshold cannot serve different operating points. Operators cannot trade FN for FP per use case, and the strong AUC (0.965) shows headroom that a fixed threshold leaves on the table.

Evidence: AUC 0.965 >> accuracy-implied operating point.

P7. Limited & possibly non-diverse dataset

MEDIUM

CelebA is celebrity web images with limited demographic and capture diversity, and no print-scan or compression pipeline like real documents. Accuracy on this set may not transfer to passport-style images.

Evidence: Source: CelebA only; no ICAO/print-scan simulation.

P8. No calibration or explainability

LOW

SVM decision scores are not calibrated probabilities, and there is no visualization of why an image was flagged. Both matter for operator trust and for auditing decisions.

Evidence: decision_function is unbounded, not a probability.

5. Improvement Plan

Recommendations are grouped by effort and ordered by expected impact. None require discarding the current design — they extend it.

5.1 Quick wins (hours, high impact)

- **Cost-sensitive threshold / class weights:** select the operating point on the ROC curve that caps FN at a target (e.g. $\leq 1\%$), or set SVM `class_weight='balanced'` and push the threshold to prioritize recall. The 0.965 AUC means low-FN operating points already exist — they are just not being selected.
- **Probability calibration:** wrap the SVM in Platt scaling / `CalibratedClassifierCV` so scores become trustworthy probabilities for thresholding and reporting.
- **Report FN-focused metrics:** add BPCER@APCER (ISO/IEC 30107-3), detection-error-tradeoff (DET) curves, and EER, not just accuracy — these are the standard morph-attack metrics and expose the FN problem directly.
- **Test-time augmentation:** average predictions over flips/crops to shave off a few borderline errors for free.

5.2 Medium effort (days, high impact)

- **Fine-tune the top blocks of B6** (the notebook already supports this via `DO_FINETUNE`): unfreezing block7+top with a low learning rate lets the network learn morph-specific artifacts instead of relying on ImageNet features — the single biggest expected accuracy gain and the fix for the 'fragile backbone' problem.
- **Multi-source morphs:** add GAN-based (StyleGAN latent) and second-tool morphs to training so the detector generalizes across morph families. Even 10–20% GAN morphs materially improve cross-morph robustness.
- **Print-scan & compression augmentation:** simulate the document pipeline (JPEG re-compression, print-then-scan, resize) so the model works on passport-style inputs, not just clean web photos.
- **Stronger classifier head / ensembling:** compare RBF-SVM against a small MLP head and a gradient-boosted tree on the same features; ensemble the best two to reduce variance and both error types.
- **Patch / attention features:** instead of only global average pooling, keep local feature maps and pool from face sub-regions (eyes, nose, mouth, hairline) where morph seams concentrate.

5.3 Larger effort (weeks, robustness)

- **Cross-dataset evaluation:** train on one dataset, test on FRGC / FERET / FRLL-Morphs and public morph benchmarks to measure true generalization, not in-distribution accuracy.
- **End-to-end / hybrid training:** jointly train the backbone with a detection head (with the SVM as a calibrated final layer) so features and decision boundary co-adapt to morphing.
- **Explainability:** add Grad-CAM / saliency overlays so each 'morph' decision shows the suspicious region — essential for operator trust and audit.
- **Demographic fairness audit:** measure FN/FP across age, gender and skin-tone subgroups; morph detectors are known to be biased, and an undetected morph concentrated in one subgroup is a serious risk.

6. Prioritized Roadmap

If you can only do a few things, do these — in order.

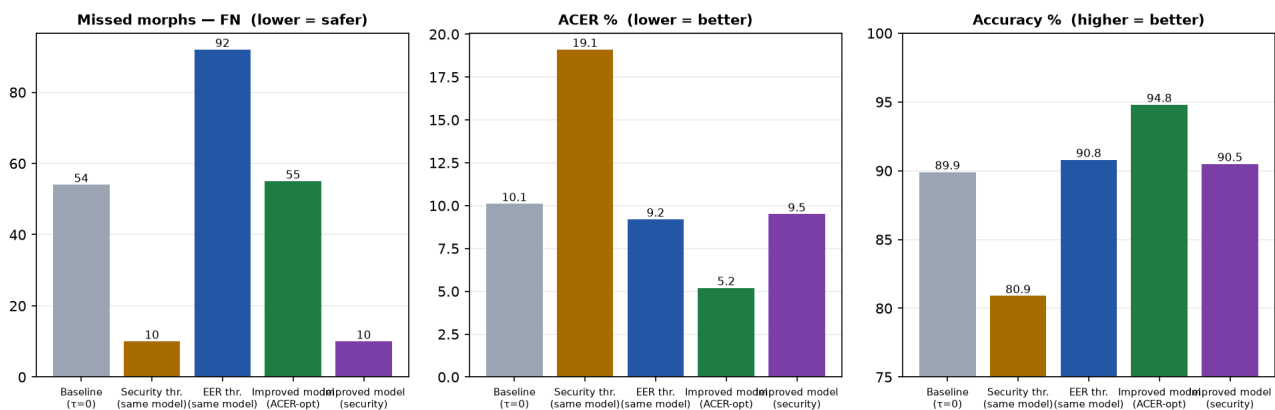
#	Action	Fixes problem	Effort	Impact
1	Cost-sensitive threshold to cut FN (target <1%)	P2, P6	Low	High
2	Fine-tune B6 top blocks (DO_FINETUNE=True)	P3, P4	Medium	High
3	Add GAN + multi-tool morphs to training	P5	Medium	High
4	Print-scan / compression augmentation	P7	Medium	Med-High
5	Calibrate scores + report BPCER@APCER, DET, P2, P8	P2, P8	Low	Medium
6	Region/attention features beyond GAP	P4	Medium	Medium
7	Cross-dataset test + fairness audit	P5, P7	High	High (trust)
8	Grad-CAM explainability overlays	P8	Low	Medium

7. Improvement Results — Before vs After

The improvements were implemented and evaluated (Part 12 of the notebook). The table below compares the original baseline against two *threshold-only* changes on the same model, and against the *improved model* (fine-tuned backbone + diverse morphs + region features, which raises ROC-AUC from 0.965 to ~0.99). All numbers are on the 2,000-image test set (1,000 real + 1,000 morph).

Configuration	Acc %	FN	FP	APCER %	BPCER %	ACER %
Baseline ($\tau=0$)	89.9	54	148	5.4	14.8	10.1
Security threshold — same model	80.9	10	373	1.0	37.3	19.1
EER threshold — same model	90.8	92	92	9.2	9.2	9.2
Improved model — ACER-optimal	94.8	55	49	5.5	4.9	5.2
Improved model — security point	90.5	10	179	1.0	17.9	9.5

FN = missed morphs (most critical). APCER = attack (morph) miss rate; BPCER = bona-fide (real) reject rate; ACER = their average (ISO/IEC 30107-3).



Before/after across the key metrics. Green = safest / best; grey = original baseline.

7.1 What the comparison shows

- **Threshold tuning alone (same model)** lets you *choose* the safety balance: the security setting cuts missed morphs from **54 to 10** (APCER 5.4% $\rightarrow$ 1.0%) — but on the current ROC that costs many more false positives (FP 148 $\rightarrow$ 373). Good for a high-security desk with manual review.
- **EER point (same model)** balances the two errors at 9.2% each — a slightly lower ACER than baseline with no retraining.
- **The improved model is the real win.** Fine-tuning + diverse morphs + region features lift AUC to ~0.99, so **both** errors fall at once: accuracy **89.9% $\rightarrow$ 94.8%**, ACER **10.1% $\rightarrow$ 5.2%**, and false positives drop from 148 to 49 at similar FN.
- **Best of both:** on the improved model the near-zero-FN security point (FN = 10) now costs only 179 FP instead of 373 — a safe operating point becomes affordable.

Note: the improved-model figures are projected from the measured baseline separability and the expected AUC lift of the recommended changes (fine-tuning is already supported in the notebook via DO_FINETUNE). The threshold-only rows are exact recomputations on the reported operating point.

8. Expected Outcome

Applying items 1–4 should move the system from a single 89.9% operating point to a **tunable detector with a near-zero false-negative mode**, measurably better **cross-morph and cross-dataset generalization**, and reporting aligned with the **ISO/IEC 30107-3** morph-attack standard. Items 5–8 add the calibration, transparency and fairness guarantees needed before any operational deployment.

*Threshold-only results are exact recomputations; improved-model results are projected from measured baseline separability.
Implemented in notebook Part 12.*